AndroidManifest.xml (Recording all activities)

```
<?xml version="1.0" encoding="utf-8"?>

<manifest xmlns:android="http://schemas.android.com/apk/res/
android"

    xmlns:tools="http://schemas.android.com/tools">


    <uses-permission android:name="android.permission.CAMERA"
/>


    <uses-feature android:name="android.hardware.camera" />

    <uses-feature
android:name="android.hardware.camera.autofocus" />


    <application

        android:allowBackup="true"

        android:dataExtractionRules="@xml/
data_extraction_rules"

        android:fullBackupContent="@xml/backup_rules"

        android:icon="@mipmap/ic_launcher"

        android:label="@string/app_name"

        android:roundIcon="@mipmap/ic_launcher_round"

        android:supportsRtl="true"

        android:theme="@style/Theme.SmartCatBox"

        tools:targetApi="31">

        <activity

            android:name=".FaceInputActivity"

            android:exported="false" />

        <activity
```

```
            android:name=".SplashActivity"
            android:exported="true">
            <intent-filter>
                <action
android:name="android.intent.action.MAIN" />

                <category
android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity> <!-- 登录页 -->
        <activity
            android:name=".MainActivity"
            android:exported="true" /> <!-- 注册页 -->
        <activity
            android:name=".RegisterActivity"
            android:exported="true" /> <!-- 人脸验证 -->
        <activity
            android:name=".FaceVerifyActivity"
            android:exported="true" /> <!-- 救助人主页 -->
        <activity
            android:name=".RescuerMainActivity"
            android:exported="true" /> <!-- 储粮查看 -->
        <activity
            android:name=".FoodStorageActivity"
            android:exported="true" /> <!-- 猫咪档案列表 -->
        <activity
            android:name=".CatListActivity"
```

```
            android:exported="true" />  <!-- 猫咪档案编辑 -->
        <activity
            android:name=".CatEditActivity"
            android:exported="true" />  <!-- 加热控制 -->
        <activity
            android:name=".HeatingActivity"
            android:exported="true" />  <!-- 用户主页（领养） -->
        <activity
            android:name=".UserMainActivity"
            android:exported="true" />  <!-- 领养详情 -->
        <activity
            android:name=".AdoptDetailActivity"
            android:exported="true" />  <!-- 错误页 -->
        <activity
            android:name=".ErrorActivity"
            android:exported="true" />
    </application>

</manifest>
```

Title(Splash) Page

activity_splash.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/
android"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="32dp"
    android:background="@drawable/splash_bg_gradient">

    <!-- Icon -->
    <ImageView
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:src="@drawable/ic_cat"
        android:layout_marginBottom="30dp"/>

    <!-- Title -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SmartCatBox"
        android:textSize="36sp"
        android:textStyle="bold"
        android:textColor="#FFFFFF"/>

    <!-- Subtitle -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Tech for Paws: A Warm Home for Strays"
```

```
        android:textSize="16sp"
        android:textColor="#F5F5F5"
        android:layout_marginTop="16dp"/>

    <!-- Enter Button -->
    <Button
        android:id="@+id/btn_enter_app"
        android:layout_width="200dp"
        android:layout_height="50dp"
        android:text="Enter"
        android:textSize="18sp"
        android:layout_marginTop="80dp"
        android:backgroundTint="#FFFFFF"
        android:textColor="#333333"/>

</LinearLayout>
```

SplashActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class SplashActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_splash)

        // Finding enter button
        val btnEnterApp =
findViewById<Button>(R.id.btn_enter_app)

        // Jump to RegisterActivity page
        btnEnterApp.setOnClickListener {
            Toast.makeText(this, "Welcome to SmartCatBox~",
Toast.LENGTH_SHORT).show()
            val intent = Intent(this,
MainActivity::class.java)
            startActivity(intent)
            finish() // Never Coming back
        }
    }
}
```

Main(Login) Page

activity_main.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="32dp"
```

```
android:background="#F5F5F5">

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login to SmartCatBox"
    android:textSize="26sp"
    android:textStyle="bold"
    android:layout_marginBottom="40dp"/>

<EditText
    android:id="@+id/etUsername"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Username"
    android:backgroundTint="#FFB88E"
    android:padding="12dp"/>

<EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    android:backgroundTint="#FFB88E"
    android:padding="12dp"
    android:layout_marginTop="16dp"/>
```

```
    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="24dp"
        android:backgroundTint="#FFB88E"
        android:textColor="#FFFFFF"/>

    <Button
        android:id="@+id/btnGoRegister"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register"
        android:layout_marginTop="16dp"
        android:backgroundTint="#FFFFFF"
        android:textColor="#FFB88E"/>

</LinearLayout>
```

MainActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
```

```
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        // Connecting Controls
        val etUsername =
findViewById<EditText>(R.id.etUsername)
        val etPassword =
findViewById<EditText>(R.id.etPassword)
        val btnLogin = findViewById<Button>(R.id.btnLogin)
        val btnGoRegister =
findViewById<Button>(R.id.btnGoRegister)

        // Register button: Jump to RegisterActivity
        btnGoRegister.setOnClickListener {
            startActivity(Intent(this,
RegisterActivity::class.java))
        }

        // Login button
        btnLogin.setOnClickListener {
            val username = etUsername.text.toString().trim()
            val password = etPassword.text.toString().trim()
```

```
            // Can't be Empty
            if (username.isEmpty() || password.isEmpty()) {
                Toast.makeText(this, "Username and Passward
Can't be Empty! ", Toast.LENGTH_SHORT).show()
                return@setOnClickListener
            }

            // Test User: Rescuer - 123  |  User - 123
            when {
                username == "Rescuer" && password == "123" ->
{
                    // Jump to FaceVerifyActivity
                    startActivity(Intent(this,
FaceVerifyActivity::class.java))
                }
                username == "User" && password == "123" -> {
                    // Jump to UserMainActivity
                    startActivity(Intent(this,
UserMainActivity::class.java))
                }
                else -> {
                    // Jump to ErrorActivity
                    startActivity(Intent(this,
ErrorActivity::class.java))
                }
            }
        }
    }
```

```
}
```

Register Page

activity_register.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="32dp"

    android:background="#F5F5F5">


    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Registration"

        android:textSize="26sp"

        android:textStyle="bold"

        android:layout_marginBottom="30dp"/>


    <EditText

        android:id="@+id/etName"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:hint="Username"

        android:backgroundTint="#FFB88E"

        android:padding="12dp"/>
```

```
<EditText
    android:id="@+id/etPwd"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Passward"
    android:inputType="textPassword"
    android:backgroundTint="#FFB88E"
    android:padding="12dp"
    android:layout_marginTop="16dp"/>

<EditText
    android:id="@+id/etAdminCode"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Admin Code (Rescuer Only) "
    android:backgroundTint="#FFB88E"
    android:padding="12dp"
    android:layout_marginTop="16dp"/>

<Button
    android:id="@+id/btnRegister"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Continue"
    android:layout_marginTop="24dp"
    android:backgroundTint="#FFB88E"
```

```
        android:textColor="#FFFFFF"/>

</LinearLayout>
```

RegisterActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class RegisterActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_register)

        // Connecting Controls
        val etName = findViewById<EditText>(R.id.etName)
        val etPwd = findViewById<EditText>(R.id.etPwd)
        val etAdminCode =
findViewById<EditText>(R.id.etAdminCode)
        val btnRegister =
findViewById<Button>(R.id.btnRegister)

        // Register Button
        btnRegister.setOnClickListener {
```

```
            val name = etName.text.toString().trim()
            val pwd = etPwd.text.toString().trim()
            val adminCode =
etAdminCode.text.toString().trim()

            // Can't be Empty
            if (name.isEmpty() || pwd.isEmpty()) {
                Toast.makeText(this, "Username and Passward
Can't be Empty! ", Toast.LENGTH_SHORT).show()
                return@setOnClickListener
            }

            // Admin Code
            when {
                adminCode == "codeadmin" -> {
                    // Jump to FaceInputActivity
                    Toast.makeText(this, "Register
Successfully! ", Toast.LENGTH_SHORT).show()
                    startActivity(Intent(this,
FaceInputActivity::class.java))
                }
                adminCode.isEmpty() -> {
                    // Jump to UserMainActivity
                    Toast.makeText(this, "Register
Successfully! ", Toast.LENGTH_SHORT).show()
                    startActivity(Intent(this,
UserMainActivity::class.java))
                }
```

```
                else -> {
                    // Jump to ErrorActivity
                    startActivity(Intent(this,
ErrorActivity::class.java))
                }
            }
        }
    }
}
```

UserMain(Cat List for User) Page

activity_user_main.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="32dp"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Cat Adoption Platform"
        android:textSize="26sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp"/>
```

```
<!-- Cat List -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:background="#FFFFFF"
    android:padding="16dp"
    android:layout_marginTop="10dp">

    <ImageView
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:src="@drawable/ic_cat"
        android:layout_marginRight="16dp"/>

    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:orientation="vertical">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Cat: Kit"
            android:textSize="18sp"
```

```
            android:textStyle="bold"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Breed: Ginger Cat | Sterilized"
            android:textSize="14sp"
            android:layout_marginTop="4dp"/>

    </LinearLayout>

    <Button
        android:id="@+id/btnAdopt"
        android:layout_width="80dp"
        android:layout_height="40dp"
        android:text="Adoption"
        android:backgroundTint="#FFB88E"
        android:textColor="#FFFFFF"/>

  </LinearLayout>

</LinearLayout>
```

UserMainActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
```

```
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity

class UserMainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_user_main)

        val btnAdopt = findViewById<Button>(R.id.btnAdopt)

        // Jump to AdoptDetailActivity
        btnAdopt.setOnClickListener {
            startActivity(Intent(this,
AdoptDetailActivity::class.java))
        }
    }
}
```

RescuerMain Page

activity_rescuer_main.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="32dp"
    android:background="#F5F5F5">
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Rescuer Page"
    android:textSize="26sp"
    android:textStyle="bold"
    android:layout_marginBottom="60dp"/>

<Button
    android:id="@+id/btnFood"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:text="Food Storage"
    android:textSize="18sp"
    android:backgroundTint="#FFB88E"
    android:textColor="#FFFFFF"
    android:layout_marginTop="16dp"/>

<Button
    android:id="@+id/btnCatList"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:text="Cat List"
    android:textSize="18sp"
    android:backgroundTint="#FFB88E"
    android:textColor="#FFFFFF"
```

```
        android:layout_marginTop="16dp"/>

    <Button
        android:id="@+id/btnHeat"
        android:layout_width="match_parent"
        android:layout_height="60dp"
        android:text="Heating Contorl"
        android:textSize="18sp"
        android:backgroundTint="#FFB88E"
        android:textColor="#FFFFFF"
        android:layout_marginTop="16dp"/>

</LinearLayout>
```

RescuerMainActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity

class RescuerMainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_rescuer_main)

        // Connecting Controls
```

```
        val btnFood = findViewById<Button>(R.id.btnFood)
        val btnCatList =
findViewById<Button>(R.id.btnCatList)
        val btnHeat = findViewById<Button>(R.id.btnHeat)

        // Jump to FoodStorageActivity
        btnFood.setOnClickListener {
            startActivity(Intent(this,
FoodStorageActivity::class.java))
        }

        // Jump to CatListActivity
        btnCatList.setOnClickListener {
            startActivity(Intent(this,
CatListActivity::class.java))
        }

        // Jump to Heating Activity
        btnHeat.setOnClickListener {
            startActivity(Intent(this,
HeatingActivity::class.java))
        }
    }
}
```

FaceInput Page

activity_face_input.xml (The Layout Displayed on the App)

```
package com.example.smartcatbox

import android.content.Intent
```

```
import android.content.SharedPreferences
import android.graphics.Bitmap
import android.os.Bundle
import android.provider.MediaStore
import android.widget.Button
import android.widget.ImageView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class FaceInputActivity : AppCompatActivity() {
    private lateinit var sp: SharedPreferences
    private val REQUEST_IMAGE_CAPTURE = 100
    private lateinit var facePreview: ImageView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_face_input)

        sp = getSharedPreferences("CatBoxConfig",
MODE_PRIVATE)
        facePreview = findViewById(R.id.face_preview)
        val btnTakeFace =
findViewById<Button>(R.id.btn_take_face)

        btnTakeFace.setOnClickListener {
            val takePicture =
Intent(MediaStore.ACTION_IMAGE_CAPTURE)
```

```
            startActivityForResult(takePicture,
REQUEST_IMAGE_CAPTURE)
        }
    }

    override fun onActivityResult(requestCode: Int,
resultCode: Int, data: Intent?) {
        super.onActivityResult(requestCode, resultCode, data)
        if (requestCode == REQUEST_IMAGE_CAPTURE &&
resultCode == RESULT_OK) {
            val imageBitmap = data?.extras?.get("data") as
Bitmap
            facePreview.setImageBitmap(imageBitmap)

            // Mark Inputted Face
            sp.edit().putBoolean("face_enabled",
true).apply()
            Toast.makeText(this, "Success",
Toast.LENGTH_SHORT).show()

            // Jump to CatEditActivity
            startActivity(Intent(this,
CatEditActivity::class.java))
            finish()
        }
    }
}
```

FaceInputActivity.kts (Codes that Makes the Layout Working)

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/
android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <ImageView
        android:id="@+id/face_preview"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:background="#eee"/>

    <Button
        android:id="@+id/btn_take_face"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Facial Input"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

FaceVerify Page

activity_face_verify.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="32dp"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Facial Test"
        android:textSize="26sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Press to Start"
        android:textSize="16sp"
        android:layout_marginBottom="30dp"/>

    <Button
        android:id="@+id/btnVerifyFace"
        android:layout_width="200dp"
        android:layout_height="50dp"
        android:text="Start"
```

```
        android:backgroundTint="#FFB88E"
        android:textColor="#FFFFFF"/>

</LinearLayout>
```

FaceVerifyActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class FaceVerifyActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_face_verify)

        val btnVerifyFace =
findViewById<Button>(R.id.btnVerifyFace)

        // Verification: Success to RescuerActivity, Fail to
ErrorActivity
        btnVerifyFace.setOnClickListener {
            // Success
            Toast.makeText(this, "Facial Verification
Successful! ", Toast.LENGTH_SHORT).show()
```

```
            startActivity(Intent(this,
RescuerMainActivity::class.java))
            finish()


            // Fail
            // startActivity(Intent(this,
ErrorActivity::class.java))
        }
    }
}
```

CatList (Cat List for Rescuer) Page

activity_cat_list.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="32dp"
    android:background="#F5F5F5">

    <Button
        android:id="@+id/btnBack"
        android:layout_width="80dp"
        android:layout_height="40dp"
        android:text="Back"
        android:backgroundTint="#FFFFFF"
        android:textColor="#FFB88E"/>
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Cat List"
    android:textSize="26sp"
    android:textStyle="bold"
    android:layout_marginTop="20dp"
    android:layout_marginBottom="30dp"/>

<!-- Test Case -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:background="#FFFFFF"
    android:padding="16dp"
    android:layout_marginTop="10dp"
    android:radius="8dp">

    <ImageView
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:src="@drawable/ic_cat"
        android:layout_marginRight="16dp"/>
```

```
<LinearLayout
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Cat: Kit"
        android:textSize="18sp"
        android:textStyle="bold"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="sterilization: Yes"
        android:textSize="14sp"
        android:layout_marginTop="4dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Appearance: 2025-10-01 18:00"
        android:textSize="14sp"
        android:layout_marginTop="4dp"/>
```

```
        </LinearLayout>

        <Button
            android:id="@+id/btnEdit"
            android:layout_width="80dp"
            android:layout_height="40dp"
            android:text="Edit"
            android:backgroundTint="#FFB88E"
            android:textColor="#FFFFFF"/>

    </LinearLayout>

</LinearLayout>
```

CatListActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity

class CatListActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_cat_list)
```

```
        // Connecting Controls
        val btnBack = findViewById<Button>(R.id.btnBack)
        val btnEdit = findViewById<Button>(R.id.btnEdit)

        // Return to RescuerMainActivity
        btnBack.setOnClickListener {
            startActivity(Intent(this,
RescuerMainActivity::class.java))
            finish()
        }

        // Jump to CatEditActvity
        btnEdit.setOnClickListener {
            startActivity(Intent(this,
CatEditActivity::class.java))
        }
    }
}
```

CatEdit Page

activity_cat_edit.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="32dp"
```

```
    android:background="#F5F5F5"
    android:scrollbars="vertical"
    android:weightSum="1">

    <Button
        android:id="@+id/btnBack"
        android:layout_width="80dp"
        android:layout_height="40dp"
        android:text="Back"
        android:backgroundTint="#FFFFFF"
        android:textColor="#FFB88E"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Editing"
        android:textSize="26sp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"
        android:layout_marginBottom="30dp"/>

    <ImageView
        android:layout_width="150dp"
        android:layout_height="150dp"
        android:src="@drawable/ic_cat"
        android:layout_gravity="center_horizontal"
        android:layout_marginBottom="30dp"/>
```

```
<EditText
    android:id="@+id/etCatName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Name (Default:N/A) "
    android:backgroundTint="#FFB88E"
    android:padding="12dp"/>

<EditText
    android:id="@+id/etCatBreed"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Breed (Default:N/A) "
    android:backgroundTint="#FFB88E"
    android:padding="12dp"
    android:layout_marginTop="16dp"/>

<EditText
    android:id="@+id/etSterilized"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Sterilization (Default:N/A) "
    android:backgroundTint="#FFB88E"
    android:padding="12dp"
    android:layout_marginTop="16dp"/>
```

```
<EditText
    android:id="@+id/etContact"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Contact Number (Default:N/A) "
    android:backgroundTint="#FFB88E"
    android:padding="12dp"
    android:layout_marginTop="16dp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginTop="40dp"
    android:paddingHorizontal="16dp">

    <Button
        android:id="@+id/btnAdopted"
        android:layout_width="120dp"
        android:layout_height="45dp"
        android:text="Adopted"
        android:textSize="14sp"
        android:backgroundTint="#FF6B35"
        android:textColor="#FFFFFF"/>

    <View
```

```
            android:layout_width="0dp"
            android:layout_height="0dp"
            android:layout_weight="1"/>

        <Button
            android:id="@+id/btnSave"
            android:layout_width="120dp"
            android:layout_height="45dp"
            android:text="Save"
            android:textSize="14sp"
            android:backgroundTint="#FFB88E"
            android:textColor="#FFFFFF"/>

    </LinearLayout>

</LinearLayout>
```

CatEditActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class CatEditActivity : AppCompatActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_cat_edit)

        // Connecting Controls
        val btnBack = findViewById<Button>(R.id.btnBack)
        val btnAdopted =
findViewById<Button>(R.id.btnAdopted)
        val btnSave = findViewById<Button>(R.id.btnSave)
        val etCatName =
findViewById<EditText>(R.id.etCatName)
        val etCatBreed =
findViewById<EditText>(R.id.etCatBreed)
        val etSterilized =
findViewById<EditText>(R.id.etSterilized)
        val etContact =
findViewById<EditText>(R.id.etContact)

        // Return to CatList
        btnBack.setOnClickListener {
            startActivity(Intent(this,
CatListActivity::class.java))
            finish()
        }

        // Adopted: Delete Document
        btnAdopted.setOnClickListener {
            Toast.makeText(this, "Cat Adopted! Document
Deleted", Toast.LENGTH_SHORT).show()
```

```
            startActivity(Intent(this,
CatListActivity::class.java))
            finish()
        }


        // Saving Information
        btnSave.setOnClickListener {
            val catName = if
(etCatName.text.toString().trim().isEmpty()) "N/A" else
etCatName.text.toString().trim()
            val catBreed = if
(etCatBreed.text.toString().trim().isEmpty()) "N/A" else
etCatBreed.text.toString().trim()
            val sterilized = if
(etSterilized.text.toString().trim().isEmpty()) "N/A" else
etSterilized.text.toString().trim()
            val contact = if
(etContact.text.toString().trim().isEmpty()) "N/A" else
etContact.text.toString().trim()


            Toast.makeText(this, "Cat Information Saved! ",
Toast.LENGTH_SHORT).show()
            startActivity(Intent(this,
CatListActivity::class.java))
            finish()
        }
    }
}
```

FoodStorage Page

activity_food_storage.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="32dp"
    android:background="#F5F5F5">

    <!-- Back Button -->
    <Button
        android:id="@+id/btnBack"
        android:layout_width="80dp"
        android:layout_height="40dp"
        android:text="Back"
        android:backgroundTint="#FFFFFF"
        android:textColor="#FFB88E"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Food Amount Percentage"
```

```
            android:textSize="26sp"
            android:textStyle="bold"
            android:layout_marginBottom="40dp"/>

        <TextView
            android:id="@+id/tvFoodPercent"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Food Storage: 100%"
            android:textSize="36sp"
            android:textColor="#FF6B35"
            android:layout_marginBottom="20dp"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="No Need for Adding"
            android:textSize="18sp"
            android:textColor="#333333"/>

    </LinearLayout>

</LinearLayout>
```

FoodStorageActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
```

```
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class FoodStorageActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_food_storage)

        // Connecting Controls
        val btnBack = findViewById<Button>(R.id.btnBack)
        val tvFoodPercent =
findViewById<TextView>(R.id.tvFoodPercent)

        // Random Percentage
        val foodPercent = 80
        tvFoodPercent.text = "当前储粮: $foodPercent%"

        // Return to RescuerMainActivity
        btnBack.setOnClickListener {
            startActivity(Intent(this,
RescuerMainActivity::class.java))
            finish()
        }
    }
}
```

Heating Page

activity_heating.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:padding="32dp"

    android:background="#F5F5F5">


    <Button

        android:id="@+id/btnBack"

        android:layout_width="80dp"

        android:layout_height="40dp"

        android:text="Back"

        android:backgroundTint="#FFFFFF"

        android:textColor="#FFB88E"/>


    <LinearLayout

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:orientation="vertical"

        android:gravity="center">


        <TextView

            android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Heating Control"
        android:textSize="26sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Temperature: 20℃"
        android:textSize="18sp"
        android:layout_marginBottom="20dp"/>

    <EditText
        android:id="@+id/etTemp"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:hint="Desired Temperature(ºC)"
        android:inputType="number"
        android:backgroundTint="#FFB88E"
        android:padding="12dp"
        android:gravity="center"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
```

```
        android:gravity="center"
        android:layout_marginTop="30dp">

        <Button
            android:id="@+id/btnStartHeat"
            android:layout_width="120dp"
            android:layout_height="50dp"
            android:text="Start"
            android:backgroundTint="#FFB88E"
            android:textColor="#FFFFFF"
            android:layout_marginRight="20dp"/>

        <Button
            android:id="@+id/btnStopHeat"
            android:layout_width="120dp"
            android:layout_height="50dp"
            android:text="Stop"
            android:backgroundTint="#FF6B35"
            android:textColor="#FFFFFF"/>

    </LinearLayout>

    <TextView
        android:id="@+id/tvHeatStatus"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Not Heating"
```

```
            android:textSize="18sp"
            android:textColor="#333333"
            android:layout_marginTop="30dp"/>

    </LinearLayout>

</LinearLayout>
```

HeatingActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class HeatingActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_heating)

        // Connecting Controls
        val btnBack = findViewById<Button>(R.id.btnBack)
        val btnStartHeat =
findViewById<Button>(R.id.btnStartHeat)
```

```
        val btnStopHeat =
findViewById<Button>(R.id.btnStopHeat)
        val etTemp = findViewById<EditText>(R.id.etTemp)
        val tvHeatStatus =
findViewById<TextView>(R.id.tvHeatStatus)


        // Return to RescuerMainActivity
        btnBack.setOnClickListener {
            startActivity(Intent(this,
RescuerMainActivity::class.java))
            finish()
        }


        // Start Heating
        btnStartHeat.setOnClickListener {
            val tempStr = etTemp.text.toString().trim()
            if (tempStr.isEmpty()) {
                Toast.makeText(this, "Enter Desired ºC! ",
Toast.LENGTH_SHORT).show()
                return@setOnClickListener
            }
            val temp = tempStr.toInt()
            tvHeatStatus.text = "Heating to: $temp℃"

tvHeatStatus.setTextColor(resources.getColor(android.R.color.
holo_red_dark))
            Toast.makeText(this, "Starting! ",
Toast.LENGTH_SHORT).show()
        }
```

```
        // Stop Heating
        btnStopHeat.setOnClickListener {
            tvHeatStatus.text = "Not Heating"

tvHeatStatus.setTextColor(resources.getColor(android.R.color.
black))
            etTemp.setText("")
            Toast.makeText(this, "Stopped! ",
Toast.LENGTH_SHORT).show()
        }
    }
}
```

AdoptDetail Page

activity_adopt_detail.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="32dp"
    android:background="#F5F5F5">

    <Button
        android:id="@+id/btnBack"
        android:layout_width="80dp"
        android:layout_height="40dp"
```

```
        android:text="Back"
        android:backgroundTint="#FFFFFF"
        android:textColor="#FFB88E"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Details"
            android:textSize="26sp"
            android:textStyle="bold"
            android:layout_marginBottom="30dp"/>

        <ImageView
            android:layout_width="200dp"
            android:layout_height="200dp"
            android:src="@drawable/ic_cat"
            android:layout_marginBottom="30dp"/>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
    android:orientation="vertical"
    android:background="#FFFFFF"
    android:padding="24dp"
    android:radius="8dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name: Kit"
        android:textSize="18sp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Breed: Ginger Cat"
        android:textSize="18sp"
        android:layout_marginTop="8dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Sterilization: Yes"
        android:textSize="18sp"
        android:layout_marginTop="8dp"/>

    <TextView
        android:layout_width="wrap_content"
```

```
            android:layout_height="wrap_content"
            android:text="Location: Beside Gym"
            android:textSize="18sp"
            android:layout_marginTop="8dp"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Contact Number: 13800138000"
            android:textSize="20sp"
            android:textColor="#FF6B35"
            android:textStyle="bold"
            android:layout_marginTop="20dp"/>

    </LinearLayout>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Please Contact for Adoption~"
        android:textSize="18sp"
        android:layout_marginTop="30dp"/>

</LinearLayout>

</LinearLayout>
```

AdoptDetailActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox


import android.content.Intent

import android.os.Bundle

import android.widget.Button

import androidx.appcompat.app.AppCompatActivity


class AdoptDetailActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_adopt_detail)


        val btnBack = findViewById<Button>(R.id.btnBack)


        // Return to UserMainActivity

        btnBack.setOnClickListener {

            startActivity(Intent(this,
UserMainActivity::class.java))

            finish()

        }

    }

}
```

Error Page

activity_error.xml (The Layout Displayed on the App)

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/
res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="32dp"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Error! "
        android:textSize="36sp"
        android:textStyle="bold"
        android:textColor="#FF6B35"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btnBackLogin"
        android:layout_width="200dp"
        android:layout_height="50dp"
        android:text="Return"
        android:backgroundTint="#FFB88E"
        android:textColor="#FFFFFF"/>

</LinearLayout>
```

ErrorActivity.kts (Codes that Makes the Layout Working)

```
package com.example.smartcatbox


import android.content.Intent

import android.os.Bundle

import android.widget.Button

import androidx.appcompat.app.AppCompatActivity


class ErrorActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_error)


        val btnBackLogin =
findViewById<Button>(R.id.btnBackLogin)


        // Return to MainActivity

        btnBackLogin.setOnClickListener {

            startActivity(Intent(this,
MainActivity::class.java))

            finish()

        }

    }

}
```